NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-100 1 August 1979
Test Operations Procedure 8-2-553
AD No. A072672

SAFETY EVALUATION - CB ITEMS

			Page
Paragraph	1.	Scope	1
	2.	Facilities and Instrumentation	1
	3.	Preparation for Test	
	4.	Test Controls	
	5.	Test Conduct	
	6.	Data Reduction and Presentation	
Appendix	A.	Equipment Operation and Support Hazard Analysis	A-1
	B.	Checklists	B-1

1. SCOPE. This TOP provides general guidance for identifying and evaluating hazards associated with CB equipment. In addition, requirements for safe test conduct are outlined. Because of the variety of equipment which may be subjected to development testing, all of the information contained in this document may not apply. Each test program must be designed to assure safety specifications are met. When specific subtests are required to evaluate the safety characteristics, the TOP for the particular type of CB equipment being evaluated will be consulted to determine which specific safety subtests are to be conducted.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

2.1.1 Chemical Facilities. Facilities designed for chemical testing will meet the requirement of either total or vapor containment depending upon the type of operation.[1] The exhaust system will be designed to prevent the discharge of toxic agents into the atmosphere in amounts harmful to humans. The 1% lethality distance for the maximum credible event will be used to site test facilities in which chemical agents are to be used.[2]

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[1]DOD 5154.4S, para 5-8h, DOD Ammunition and Explosives Standards, Jan 78.
[2]DARCOM-R 385-102, para 15-2, Safety Regulations for Chemical Agents GB and VX, 28 Feb 77.

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2.1.2 Biological Facilities. Facilities designed for biological testing will be designed to provide the degree of containment specified in Classification of Etiologic Agents on the Basis of Hazard[3] for the type biological material being used.

2.2 Instrumentation. Each work area must be monitored to insure that hazardous concentrations of chemical agents or biological pathogens are not present. The lower detection limit of the technique used must be the agent levels established by the Surgeon General for unprotected personnel. For chemical agents the levels are 0.001 mg/m^3 for GB (1 hour),[4] 0.005 mg/m^3 for VX (1 hour),[5] and 0.01 mg/m^3 for HD (3 hours).[6]

3. PREPARATION FOR TEST.

3.1 Preliminary Safety Review and Documentation Preparation.

3.1.1 Ensure the presence of a safety statement from the developer.[7] All developer or contractor identified hazards should be documented in the safety statement. All hazards identified must be taken into account in test planning.

3.1.2 Ensure that specific tests are included in the test plan to verify compliance with safety criteria established for the system. These tests will differ for the different types of equipment being tested and are usually described in the TOP for tne specific item being tested.

3.1.3 Review the system support package, all instructional material, literature and draft technical manuals.

3.1.4 Ensure that site plan, hazard zone calcuations and safety submissions required by DOD 5154.4S[8] and DARCOM-R 385-100[9] have been submitted for chemical operations or CB facility siting.

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[3]Classification of Etiologic Agents on the Bais of Hazard, US Department of Health, Education and Welfare Center for Disease Control, Sep 76.
[4]DARCOM-R 385-102, para 3-1a(3)(b), Safety Regulations for Chemical Agents GB and VX, 28 Feb 77.
[5]Ibid, para 3-1b(3)(b).
[6]DARCOM-R 385-31, para 2-4, Safety Regulations for Chemical Agents H, HD and HT, 20 Apr 79.
[7]AR 70-10, para 2-21b(4), Test and Evluation During Development and Acquisition of Material, 29 Aug 75.
[8]DOD 5154.4S, para 5-8h, DOD Ammunition and Explosives Standards, Jan 78.
[9]DARCOM-R 385-102, para 6-5, Safety Regulations for Chemical Agents GB and VX, 28 Feb 77.

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3.1.5 All plans for open-air testing of chemical agents must comply with the requirements of Section 409, Public Law 91-121 and Section 506, Public Law 91-441, and have prior review by the Surgeon General (TSG). Environmental Impact Statements (EIS) must be prepared and published as required by AR 200-1.[10]

3.1.6 If required by AR 70-25, an experimental protocol with consent statement will be submitted to the Surgeon General (TSG) for approval. In general, safety and human factors determinations in developmental testing should be limited to verifying that previously approved exposure limits are not exceeded. Any human experimentation necessary to answer basic questions pertaining to safety, health and stress hazards should have been completed before development testing. TECOMR 385-7[11] describes policy for human participation in testing. It provides procedures for insuring that no test participant is exposed to health hazards beyond normal duty limits.

3.1.7 Ensure required Standing Operating Procedures and Test Operation Procedures are prepared. The procedures for all hazardous operations will be approved by the installation or test activity commander or his designated representative.[12]

3.2 Operator Training and Familiarization.

3.2.1 Ensure that required new equipment operator training is conducted by the developer.

3.2.2 Conduct a preoperational briefing for all personnel prior to the start of the test. All personnel will review the hazards and precautions outlined in the Test Operation Procedure and Safety Annex.

3.2.3 Conduct an Operational Readiness Inspection (ORI) prior to conduct of any test involving lethal or incapacitating chemical agents.

3.3 Receipt Inspection. An initial safety inspection of the test item shall be performed by qualified personnel prior to conducting all other tests of the item. The checklists in Appendix B provide a guide to the types of hazards which may be encountered. Satisfactory resolution shall be obtained concerning all potential hazards prior to conducting tests.

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[10]AR 200-1, Environmental Protection and Enhancement, 20 Jan 78.
[11]TECOMR 385-7, Potential Health Hazards to Humans Participating in Testing, 15 Nov 78.
[12]DARCOM-R 385-100, para 16-21, Safety Manual, 21 Apr 70.

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4. TEST CONTROLS.

4.1 Procedures for Accumulating Data. The results of hazard analysis, accident investigations, inspections, interviews and operator reports of unsafe conditions will be used as methods of accumulating data.

4.2 Personnel. Test Officer, Project Officer and an occupationally qualified Safety Officer will participate in the accumulation of data and the analysis of the data using risk assessment procedures.

4.3 Risk Assessment. Risk assessment procedures will be used to establish the degree of hazard. The probability that the mishap will occur and the severity of potential consequences will be considered in this evaluation. Risk assessment procedures are described in Appendix C.

5. TEST CONDUCT.

5.1 Safety Subtests. Specific safety subtests will be designed to evaluate all safety criteria established for an item or to otherwise determine safety or maintenance hazards. The subtests are usually described in the TOP for the item being tested. The following is an overview of each type of equipment and the types of tests which should be conducted.

5.1.1 Auxiliary Equipment (CB). These items are usually used in conjunction with other items during shipment, filling, impregnation of clothing. Shipping containers must meet Department of Transportation[13] requirements and material handling devices must meet Occupational Safety and Health Act (OSHA) standards.[14]

5.1.2 Collective Protection Systems (CB). These systems provide and maintain filtered air under positive pressure at a level which prevents the infiltration of hazardous levels of CB agents. Included in this category are: vehicles and vans, field shelters, and fixed installations. The principal safety consideration is to ensure that an adequate amount of contamination-free air is available to accommodate personnel working in the shelter. Means must be provided to prevent contamination via personnel exit/entry. Decontamination of the system and electrical/mechanical hazards associated with generators and other anciliary equipment must be considered. Heat stress, fatigue and noise may be contributing factors.

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[13] 49CFR parts 171-177, Hazardous Materials Regulations, US Department of Transportation, Materials Transportation Bureau.
[14] 29CFR para 1910, General Industry Safety and Health Standards, US Department of Labor Occupational Safety and Health Administration.

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Equipment assembly/disassembly procedures should be tested. MIL-HNBK-753[15] contains descriptive information on CB collective protection systems.

5.1.3 Decontamination Equipment (CB). The items include the various processes of absorbing, destroying, neutralizing or removing chemical or biological agents to make any contaminated person, object or area safe. Included in this category are portable decontaminating apparatus, power-driven vehicle or skid-mounted decontaminating apparatus and individual decontamination kits. Information on decontamination is found in TOP 8-2-510.[16] Safety considerations should include the mechanical and electrical hazards associated with generators, mixing paddles, pressure hoses, etc. Consideration must be given to the toxicological effects of each chemical used as well as the reaction with toxic chemical agents.

5.1.4 Detection and Surveillance Equipment (CB). This equipment is intended to provide notice that CB agent is present and protective measures must be taken or continued. This classification of equipment includes: biological and chemical alarms, detector kits, sampling and analyzing kits and personnel detectors. This equipment may use electronic circuitry, chemical indicators (chemical change), or electromagnetic radiation to detect the presence of chemical agents. Sensitivity, response time and false alarm rate are test criteria which the test plan must address. Consideration should be given to electrical,[17] chemical and biological hazards in addition to human factors[18] requirements.

5.1.5 Dissemination Devices (C). This equipment is intended for use in distributing relatively large quantities of chemical agents over a large area. These items include portable, vehicular or helicopter-mounted riot control agent dispersers, aircraft screening smoke systems, spray tanks, mechanical smoke generators and flame throwers. Consideration should be given to mechanical, electrical, chemical, fire and procedural hazards associated with this equipment. Additional information is contained in TOP 8-3-80.[19]

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[15]MIL Handbook 753, CB Collective Protection of Mobile Vehicles, Vans and Shelters, 31 Aug 71.
[16] TOP 8-2-510, Decontamination, 30 Sep 67.
[17] MIL-STD-454F, Standard General Requirements for Electronic Equipment, 15 Mar 78.
[18] MIL-STD-1472B, Change 2, Human Engineering Design Criteria for Military Systems, Equipment and Facilities, 10 May 78.
[19]TOP 8-3-80, Airborne Dissemination Devices, 5 Mar 71.

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5.1.6 Munitions (C). These items are designed for the delivery of chemical agents, smoke or incendiaries to selected targets, and include: chemical bomblets, smoke pots, hand or weapon-launched bomblets, land mines, projectiles, cartridges, and rockets and guided missile warheads. Design criteria for chemical munitions must include provisiosn to (a) prevent release of chemical agent when the munition is involved in an accident or is jettisoned, (b) prevent inadvertent arming, launching or firing, and (c) prevent personnel exposure to chemical agent during manufacture, shipment and storage. Safety tests must develop data to insure safe storage, handling, transportation, launch and flight, and when applicable, confirm proposed surface danger areas and downwind hazard areas.[20] Tests for launch and flight safety are described in TOP 4-2-503[21] and 4-2-504.[22] Environmental test methods are described in MIL-STD-810C.[23] Hazard classification and storage compatibility tests must be conducted IAW TB 700-2.[24] Munition handling, render safe, demilitarization and disposal procedures should be verified by test.

5.1.7 Individual Protective Equipment (CB). These items are designed to protect the individual from the effects of CB agents by (1) preventing the entry of such agents into the respiratory system and (2) decreasing the cutaneous and percutaneous hazards. Included in this category are: breathing apparatus, respirators, protective suits, hoods, gloves, liners, masks, overgarments and casualty bags.

Federal respiratory testing and certification procedures[25] will be used as guides in establishing filter test programs. Protective mask filters will meet the criteria of a Type N filter; however, representative chemical agents will be used in the testing. Human engineering aspects are of paramount concern. The following factors must be evaluated: (1) comfort and fit while wearing, (2) ease of breathing (inhalation and exhalation), (3) visibility, (4) interference with other equipment carried or used, (5) interference with normal duties, (6) ease of donning, (7) heat stress, (8) speech transmission, (9) skin irritation, (10) flammability,

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[20] DARCOM-R 385-24, Development of Army Range Safety Regulations, 11 Apr 74.
[21] TOP 4-2-503, Safety Evaluation of Close Support Rockets and Missiles, 1 Jul 71.
[22] TOP 4-2-504, Safety Evaluation - Artillery, Motor and Recoilless Rifle Ammunition, 11 Dec 72.
[23] MIL-STD-810C, Environmental Test Methods, 10 Mar 75.
[24] TB 700-2, Explosive Hazard Classification Procedures, 19 May 67 with 1 Change.
[25] Code of Federal Regulations, Title 30, Mineral Resources, Part 11 (30CFR11).

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(11) decontamination, and (12) disposal procedures. Test methods are described in MIL-STD 282[26] and FED-STD-191.[27]

5.2 Equipment Operation and Support Hazard Analysis. The Equipment Operation and Support Hazard Analysis is conducted in conjunction with other subtests. The following hazards should be evaluated using the technique described in Appendix A:

Mechanical Hazards

Electrical Hazards

Chemical Hazards

Biological Hazards

Physiological Hazards

Fire and Explosion Hazards

Procedural Hazards (operating and maintenance)

5.2.1 Mechanical Hazards.

5.2.1.1 Carefully examine all instructional material to determine potential mechanical hazards.

5.2.1.2 Accomplish a thorough test item inspection and observe the item throughout all testing and evaluations.

5.2.1.3 The following potential mechanical hazard sources should be considered when performing this evaluation:

Rotating, reciprocating and transverse motions

Cam Action

Cutting Actions - Motion

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[26]MIL-STD-282, Change 2, Filter Units, Protective Clothing, Gas Mask Components and Related Products, Performance Test Method, 10 Jun 74.
[27]FED-STD-191, Change 5, Textile Test Methods, 17 Jul 74.

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Cutting exposure - sharpness

Punching, shearing and bending actions

Rate of speed

Instability (center of gravity)

Entrapment

Lack of clearance

Misleading appearance of quality

Stored energy - physical and electrical

Improper rigidity

Impact

5.2.1.4 A sample checklist is included in Appendix B.

5.2.2 Electrical Hazards.

5.2.2.1 Examine all instructional material; determine the location of all potential electrical hazards and ensure that these hazards are clearly indicated and that appropriate caution notices and instructions are provided.

5.2.2.2 The test item shall be thoroughly inspected and observations made throughout conduct of all tests and evaluations.

5.2.2.3 The following electrical hazard sources should be considered when performing this evaluation:

Shock

Short circuit

Stored electrical charge (batteries and capacitors)

Stray voltage

Improper and/or inadequate ground

Fire

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Overheating

Ventilation

Insulation failure

Radiation - X-ray

Sparks

Arcing

Explosion

5.2.2.4 A sample checklist is included in Appendix B.

5.2.3 Biological Hazards.

5.2.3.1 Determine the classification or precaution category for each microorganism which may be associated with this equipment.[28]

5.2.3.2 Ensure that operational requirements and the level of competence for personnel using the equipment have been established. Specific tests in the safety subtest will be conducted to ensure requirements have been met.

5.2.3.3 Secure facility to preclude entry of nonimmunized visitors in street clothing. Post "BIOHAZARD" warning signs.

5.2.3.4 The following factors require evaluation for biological hazards and should be reviewed for their influence on the system:

Immunization availability and requirements

Classification of the agent on the basis of hazard

Availability of recommended prophylaxis and decontaminating measures

Special protective clothing

Work involving insects/animals

Aerosol precautions to avoid exposure

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[28]Classification of Etiological Agents on the Basis of Hazard, Center for Disease Control, 1976, 13pp.

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5.2.3.5 A sample checklist is included in Appendix B.

5.2.4 Chemical Hazards.

5.2.4.1 Determine each chemical contained in or used with this equipment.

5.2.4.2 When exposure of personnel to chemicals will occur during operation of the system, ensure that the health hazards for each chemical have been considered and that sampling is conducted to insure that exposure limits are not exceeded. If protective devices are used to eliminate the exposure their adequacy must be evaluated.

5.2.4.3 In addition, each chemical should be reviewed for the following properties and their affect on the system:

Corrosive

To animals

To plants

To materials

Toxic

By inhalation

By skin absorption

By ingestion

Flammable

Pyrophoric

Explosive

Shock sensitive

Oxidizing

Photoreactive

Water reactive

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Peroxide former

Carcinogenic

Teratogenic

Decomposes

5.2.4.4 A sample checklist is included in Appendix B.

5.2.5 Physiological Hazards.

5.2.5.1 Through the conduct of the test, note any physiological hazards which might be hazardous to the operation or maintenance personnel. Specific industrial hygiene measurements should be made to verify suspected hazards.

5.2.5.2 The following are sources of physiological hazards which may be considered:

Noise (pressure)

High intensity

High frequency

Impulsive

Vibration

Radiation, ionizing

Radiation, nonionizing

Ultra Violet emission

Visible light emission

Infrared emission

Microwave emission

Radio wave emission

Physical stress

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Temperature - variation - extremes

Oxygen depletion

Lifting and carrying

5.2.5.3 A sample checklist is contained in Appendix B.

5.2.6 Fire and Explosion Hazards.

5.2.6.1 Accomplish a thorough test item inspection and observe the item throughout all testing and evaluations for fire and explosion hazards. The following fire and explosion related factors should be considered:

Fuel source

Rate of flammability

Ignition source

Heat (chemical)

Heat (spontaneous)

Heat (mechanical)

Spark (mechanical)

Spark (electrical - static)

Open flame

5.2.6.2 A sample checklist is included in Appendix B.

5.2.7 Procedure Hazards. The primary variables to be considered are the operator, his surrounding environment and the tasks which he must perform. Each task must be examined to determine the danger to personnel and equipment.

6. DATA REDUCTION AND PRESENTATION.

6.1 Data Presentation. The format shown in Figure 1, Appendix A will be used to present all conditions which are hazardous to personnel, equipment and property.

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6.2 Narrative Description of Test Results. Sufficient narrative comments will be included on each condition to provide background information to be used in the analysis of test results.

6.3 Analysis. Each hazardous condition will be analyzed as outlined in Appendix A to determine the category (severity) and probability of the hazard. The classification guide in TOP 1-1-012[29] will be used to classify deficiencies, shortcomings and suggested improvements.[30]

Recommended changes to this publication should be forwarded to Commander, US Army Test and Evaluation Command, ATTN: DRSTE-AD-M, Aberdeen Proving Ground, MD 21005. Technical information related to this publication may be obtained from the preparing agency, Commander, US Army Dugway Proving Ground, ATTN: STEDP-SA, Dugway, UT 84022. Additional copies of this document are available from the Defense Documentation Center, Cameron Station, Alexandria, VA 22314. This document is identified by the accession number (AD No.) printed on the first page.

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[29]TOP 1-1-012, Classification of Deficiencies and Shortcomings.
[30]DARCOM-R 385-12, Life Cycle Verification of Materiel Safety, 29 Jun 72.

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APPENDIX A

EQUIPMENT OPERATION AND SUPPORT HAZARD ANALYSIS

Operating and support hazard analysis may be performed during development testing to identify hazards associated with equipment, procedures and personnel. Data from all development test subjects can be used to prepare the Operation and Support Hazard Analysis.

Figure 1 is a typical Operation and Support Hazard Analysis Worksheet. The following instructions apply to the worksheet:

a. Operation - What operation is being performed?

b. What function does the operation perform?

c. Effect - Normal: What is the normal effect of the function?

d. Effect - Hazard: What is the hazard which can result from the operation?

e. Hazard Category:[31]

CATEGORY I - CATASTROPHIC: May cause death or system loss

CATEGORY II - CRITICAL: May cause severe injury, severe occupational illness or major system damage

CATEGORY III - MARGINAL: May cause minor injury, minor occupational illness or minor system damage

CATEGORY IV - NEGLIGIBLE: Will not result in injury, occupational illness or system damage

f. Hazard Effect Frequency - What is the qualitative probability that the hazard effect will occur?

HAZARD PROBABILITY

Descriptive Word	Level	Specific Individual Item	Inventory
Frequent	A	Likely to occur frequently	Continuously experienced

[31]MIL-STD-882A, Systems Safety Program Requirements, 28 Jun 77

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Descriptive Word	Level	Specific Individual Item	Inventory
Reasonably Probable	B	Will occur several times in life cycle of an item	Will occur frequently
Occasional	C	Likely to occur sometime in life cycle of an item	Will occur several times
Remote	D	So unlikely it can be assumed that this hazard will not be experienced	Unlikely to occur but possible
Extremely Improbable	E	Probability of occurrence cannot be distinguished from zero	So unlikely it can be assumed it will not be experienced
Impossible	F	Physically impossible to occur	Physically impossible to occur

g. Comments relative to what should be done to prevent the hazard or protect against the consequences.

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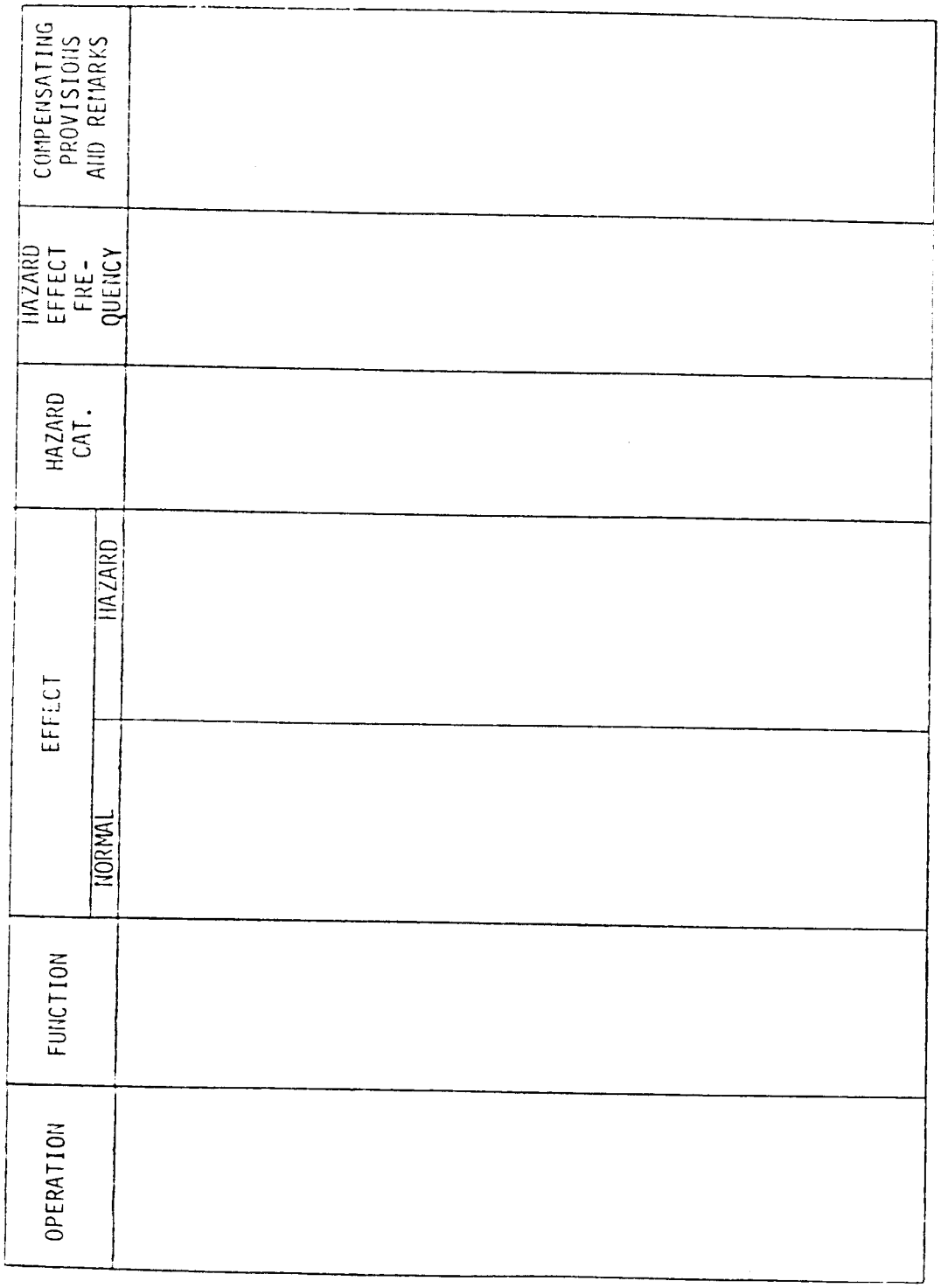


Figure 1. TYPICAL OPERATIONAL AND SUPPORT HAZARD ANALYSIS WORKSHEET

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		HAZARD PROBABILITY					
		FREQUENT	REASONABLY PROBABLE	OCCASIONAL	REMOTE	EXTREMELY IMPROBABLE	IMPOSSIBLE
SPECIFIC INDIVIDUAL ITEM →		Likely to occur frequently	Will occur several times in life of item	Likely to occur sometime in the life of item	So unlikely, can be assumed that this hazard will not be experienced	Probability of occurrence cannot be distinguished from zero	Physically impossible to occur
FLEET OR INVENTORY →		Continuously experienced	Will occur frequently	Will occur several times	Unlikely to occur, but possible	So unlikely, can be assumed that this hazard will not be experienced	Physically impossible to occur
		A	B	C	D	E	F
HAZARD SEVERITY							
CATASTROPHIC - May cause death or system loss	I	DEFICIENCY	DEFICIENCY	DEFICIENCY	DEFICIENCY	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
CRITICAL - May cause severe injury or illness, or major system damage	II	DEFICIENCY	DEFICIENCY	DEFICIENCY	SHORTCOMING	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
MARGINAL - May cause minor injury or illness, or minor system damage	III	DEFICIENCY	SHORTCOMING	SHORTCOMING	SUGGESTED IMPROVEMENT	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
NEGLIGIBLE - Will not result in injury or illness, or system damage	IV	SHORTCOMING	SUGGESTED IMPROVEMENT	SUGGESTED IMPROVEMENT	SUGGESTED IMPROVEMENT OR ACCEPTABLE	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE

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APPENDIX B

CHECKLISTS

I. MECHANICAL HAZARD CHECKLIST

This checklist may be used as a guide when testing CB items of equipment which are predominantly mechanical in nature. Examples of these items are collective protection systems, power driven decontaminating apparatus and smoke generators.

YES	NO	NA	
			1. Is the equipment designed so that the center of gravity, configuration or location of legs and supports make the equipment unlikely to tip over from unbalance effects or strong wind?
			2. Are expandable and collapsible structures such as shelters, jacks, supports, masts, tripods, etc., free from projections, sharp edges or design features which might be hazardous to personnel or associated equipment?
			3. Are lifting rings or slings provided for equipment which is normally moved or lifted by machine?
			4. Are ladders, climbing rings, handholds, rails, walkways, etc., provided where needed?
			5. Are steps and ladders and methods of supporting them safely made?
			6. Are entrances to equipment shelters free of hazardous obstructions?
			7. Do floor surfaces provide adequate nonslip characteristics?
			8. Are fasteners and methods of securing equipment to walls and racks sufficiently strong to prevent breakaway and falling?
			9. Can equipment shelters mounted on vehicles be entered without encountering a hazard?
			10. Does the installation of equipment on vehicles provide sufficient mechanical strength to minimize potential safety hazards?

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YES	NO	NA	
			11. Are provisions made in vehicular and shelter installations for securing equipment, tools and accessories during movement?
			12. Are safety measures provided in the event the trailer becomes detached from the towing vehicle?
			13. When semi-trailers are detached from towing vehicles do dolly wheels or landing gear provide adequate support?
			14. If a standard military vehicle has been modified to accommodate the equipment, is the vehicle still capable of satisfactory and safe operation?
			15. Do doors and hinged covers have positive action hold-open devices?
			16. Are locking mechanisms for doors and drawers designed to prevent injury to the operator when the lock is released?
			17. Are limit stops provided on roll-out racks and drawers?
			18. Are there provisions for easily overriding limit stops on roll-out racks and drawers?
			19. Is the method of opening a cover evident from the construction of the cover? If not, is an instruction plate permanently attached to the outside of the cover?
			20. Is it evident when a cover is in place but not secured?
			21. Is the equipment provided with suitable carrying handles?
			22. Are handles recessed rather than extended where they might be hazardous?
			23. Are handles positioned so they cannot catch on other units, wiring, or protrusions?
			24. Are handles located over center of gravity whenever possible?
			25. Are doors and other openings free of hazards from improperly designed catches, hinges, supports, fasteners and stops?
			26. Are components placed to allow sufficient space for use of test equipment and tools?

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YES	NO	NA	
			27. Are heavy parts located as close as possible to load bearing structures and as low as possible?
			28. Is the weight distribution such that the equipment is easy to handle, move or position?
			29. Are tasks of operation and maintenance such that they do not require excessive physical strength?
			30. When the equipment is to be manpacked is the weight and configuration such that the combat effectiveness of the test soldier is not jeopardized?
			31. Is the equipment free of sharp or overhanging edges and corners that might cause injury to personnel?
			32. When glass is used is it glareproof and shatterproof?
			33. Do exposed gears, cams, levers, fans, belts or other reciprocating, rotating or moving parts have adequate safety covers?
			34. When required are provisions made for protection against eye hazards from flying particles?
			35. Are safety valves, relief valves or other safety devices adjusted to their proper setting?
			36. Are potential mechanical hazards adequately treated in the instructional manual?

II. ELECTRICAL HAZARD CHECKLIST

This checklist may be used as a guide when testing CB items of equipment employing electronic circuitry or generated electricity. Chemical and biological alarms are in this category.

YES	NO	NA	
			1. Is the path to ground from the equipment continuous and permanent?
			2. Does the grounding system have sufficient mechanical strength to minimize the possibility of accidental ground disconnection?
			3. Is the ground connection to the chassis or frame mechanically secured by one of the following methods?

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YES	NO	NA	
			a. Secured to a spot welded terminal lug. b. Secured to a portion of the chassis or frame that has been formed into a soldering lug. c. Secured by a screw or nut and a lockwasher to a terminal on the ground wire.
			4. Is the grounding system of sufficient gauge size to conduct safely any currents that may be imposed upon it?
			5. Is the impedance of the ground system sufficiently low to limit the potential aboveground and to facilitate the operation of the overcurrent devices in the circuits?
			6. Are ground connections to shields and other mechanical parts, except the chassis and frame, made independently of the electrical circuits?
			7. Do plugs and convenience outlets for use with portable tools and equipment have provisions for automatic grounding?
			8. Are all external metal parts, control shafts, bushings and shields at ground potential at all times?
			9. Are voltages properly marked?
			10. Are guards, safety covers and warning plates provided for items handling high voltages?
			11. Are built-in test points provided where measurements endangering life must be made?
			12. Can high voltage circuits and capacitors be discharged to 30 volts within 2 seconds or less by automatic protective devices?
			13. When equipment is designed to operate on more than one type input power, are adequate precautions taken to prevent connection of improper power?
			14. Are DC power connections clearly marked for polarity?
			15. Are adjustment screws or other commonly worked on parts located away from unprotected high voltages?

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YES	NO	NA	
			16. Are tools to be used near high voltages adequately insulated?
			17. Do meters have protection against high voltage or current at the terminals?
			18. Are compartments where access is required for adjustment purposes equipped with interlocks with by-pass devices which remove all potentials in excess of 30 volts root mean square?
			19. In compartments where access into the interior is required for adjustment purposes and no interlocks are used, are voltages in excess of 30 volts root mean square isolated with barriers or guards?
			20. Are mechanical and electrical interlocks designed to prevent energizing by movement when men are in positions where it could be dangerous?
			21. Are internal controls located at safe distances from dangerous voltages?
			22. Are physically similar but electrically noninterchangeable components keyed so that it is impossible to insert a wrong unit?
			23. Where design considerations require plugs and receptacles of similar configuration, are mating plugs and receptacles suitably coded and marked?
			24. Is shielding sufficiently separated from exposed conductors to prevent shorting or arcing?
			25. Are wires and cables adequately supported and terminated to prevent shock and fire hazard?
			26. Are wires and cables properly protected at points where they pass through metal partitions?
			27. Can maintenance be accomplished with shielding in place?
			28. Do floor surfaces provide adequate insulating characteristics?
			29. Are emergency controls placed in readily accessible positions?

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YES	NO	NA	
			30. Is the main power breaker in an easily accessible location?
			31. Does the main power breaker cut off all power to the complete equipment or system?
			32. Can the power be cut off while installing, replacing or interchanging a complete equipment, assembly or part thereof?
			33. Are safety switches provided which will deactivate associated mechanical drive units without disconnecting other parts of the equipment?
			34. Are remotely located assemblies provided with safety switches to allow independent disconnectin of the equipment?
			35. Are potential electrical hazards adequately treated in the instruction manual?
			36. Are disconnect devices (circuit breakers) properly labeled?

III. CHEMICAL HAZARD CHECKLIST

This checklist may be used as a guide when testing equipment which uses chemicals such as chemical detectors and alarms, decontamination and collective protection systems.

YES	NO	NA	
			1. Has each chemical used in or with the system been identified in the safety statement?
			2. Have approved time-concentration exposure limits been established for each chemical used? If not, are toxicity tests being performed and interim safety precautions provided by the Surgeon General?
			3. Has each condition necessary for exposure to personnel or release to atmosphere or water been evaluated?
			4. Are the time-concentration exposure limits to personnel exceeded during operation of the item?
			5. Are precautions made to prevent exposure to respiratory hazards adequate? Skin absorption? Ingestion?
			6. Have all possible chemical reactions between the materials involved been analyzed including those with materials used in conjunction with the item being tested?

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YES	NO	NA	
			7. Are operator means of detecting a hazardous condition adequate?
			8. Are all harmful chemicals properly identified with appropriate caution notices?
			9. Are adequate safety devices and safety instructions provided for handling and use of gases stored under high pressure and/or extremely low temperature?
			10. Has the effect of decontamination procedures on the equipment surface been studied? Is chemical or biological material retained in the paint or material? What is the desorption rate?
			11. Did any personnel suffer irritation dermatitis as a result of contact with the chemical materials?
			12. Are air intakes isolated from the exhaust?
			13. Are adequate oxygen levels maintained inside shelters, etc.?
			14. Is the collective efficiency of material collection equipment (scrubbers, filters, incinerators) adequate to prevent hazardous conditions?
			15. Are the safeguards in event of power outage adequate?
			16. Are adequate disposal procedures provided for all chemicals used as a part of or with the item?

IV. BIOLOGICAL HAZARD CHECKLIST

This checklist may be used as a guide when testing equipment which uses biological agents such as detectors and alarms, decontamination and collective protection systems.

YES	NO	NA	
			1. Does the facility design provide for:
			a. Isolated work area?
			b. Controlled access?
			c. Negative air pressure?
			d. Bacteriological hoods and cabinets with adequate linear air exhaust?

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YES	NO	NA	
			2. Has a compatible method suitable for disinfection of work surfaces, work areas and equipment been established?
			3. Have protective clothing requirements been established?
			4. Are containment facilities available for insect and animal vectors?
			5. Are separate isolation chambers provided for centrifuges, blenders and other equipment capable of producing infectious aerosols?
			6. Have effective, safe, licensed immunizations been developed? Are they required?
			7. Are autoclaves and ovens for sterilization operating at sufficient temperature and pressure?
			8. Have methods for disposal of biological waste been provided?
			9. Have adequate procedures been provided for packaging?

V. PHYSIOLOGICAL HAZARDS CHECKLIST

YES	NO	NA	
			1. Is the ambient noise level acceptable for personnel safety and efficiency?
			2. Have all physical operator stresses such as repetitive motions, awkward working conditions and vibration been evaluated?
			3. Have all mental demands on operators been evaluated?
			4. Have all lifting and carrying requirements been evaluated?
			5. When necessary, have all ear and eye protective devices been provided?
			6. Are adequate controls included to prevent exposure in excess of standards to ionizing radiation?
			7. Are adequate controls included to prevent exposure in excess of standards to nonionizing radiation, including UV, IR, laser and microwave?
			8. Are adequate illumination levels available for the tasks required?
			9. Has heat stress to personnel as the result of exposure to high temperature or wearing protective equipment been evaluated?

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VI. FIRE AND EXPLOSION HAZARD CHECKLIST

YES	NO	NA	
			1. Have all possible ignition sources been evaluated to determine potential hazard?
			2. Has the flammability of the materials been taken into account in planning for use of the item?
			3. Are fire extinguishers of the proper type for the equipment provided and mounted in easily accessible locations?
			4. Are properly marked fire exits provided in shelters when required?
			5. Have precautions been taken to assure that the storage and distribution of flammable material is done safely?
			6. Is a self-closing metal can provided for oily rags and waste where required?
			7. Have fire extinguishing methods been included in technical publications?

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This TOP describes development test procedures required to determine whether chemical/biological (CB) equipment are free from design, operational or support hazards which could prevent accomplishment of their intended mission. Checklists and hazard analysis formats areprovided to assist test personnel in the assessment of hazards.

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